

The Association between Glaucoma Prevalence and Supplementation with the Oxidants Calcium and Iron

Purpose. To investigate the relationship between supplementary consumption of the oxidants calcium and iron and the prevalence of glaucoma. **Methods.** This cross-sectional study included 3833 participants in the National Health and Nutrition Examination Survey (NHANES) for 2007 and 2008, ≥ 40 years of age, who reported a presence or absence of glaucoma. Participants were interviewed regarding the use of dietary supplements and antacids during the preceding 30-day period. Data pertaining to the supplementary intake of calcium and iron was aggregated and divided into quintiles. Information regarding the presence or absence of glaucoma and demographics, comorbidities, and health-related behavior was obtained via interview. **Results.** Participants who consumed ≥ 800 mg/d of supplementary calcium or ≥ 18 mg/d of supplementary iron had significantly higher odds of having been diagnosed with glaucoma than did those who had not consumed supplementary calcium or iron, after adjustment for potential confounders (odds ratio [OR] 2.44, 95% confidence interval [CI] 1.25–4.76 for calcium; OR 3.80, 95% CI 1.79–8.06 for iron). Concurrent consumption of both calcium and iron above these levels was associated with still greater odds of having been diagnosed with glaucoma (OR 7.24, 95% CI 2.42–21.62). A clear dose–response relationship between quintiles of supplementary calcium or iron intake and glaucoma prevalence was not found. **Conclusions.** These results suggest that there may be a threshold intake of iron and calcium above which there is an increased risk of development of glaucoma. Prospective longitudinal studies are needed, to assess whether oxidant intake is a risk factor for development and progression of glaucoma.